

The Cognitive Architecture of Bold Design: A 2025 Deep Dive into Brutalism, Gaze Mechanics, and Interactive Immersion

Executive Summary: The Paradigm Shift from Efficiency to Expression

The digital landscape of 2024 and 2025 is witnessing a profound departure from the sanitized, homogenized "Corporate Memphis" aesthetic that dominated the previous decade. We are observing a pivot toward "Modern Web Brutalism" and Avant-Garde layouts—design philosophies that prioritize raw expression, high contrast, and structural non-conformity. This report provides an exhaustive analysis of the psychological underpinnings and performance metrics of these bold design strategies.

Contrary to the long-held belief that deviation from standard grids destroys usability, recent data suggests that when executed with intent, "bold" design can leverage the **Aesthetic-Usability Effect** to mask interaction costs and significantly deepen engagement. Our analysis of heatmap studies and gaze plots reveals that while traditional F-patterns are disrupting, they are often replaced by "Pinball Patterns" in broken grids or strictly linear progressions in scrollytelling, both of which offer distinct opportunities for directing user attention. Furthermore, the integration of 3D elements (Spline/Three.js) is shifting from novelty to necessity in high-end commerce, driving conversion lifts of up to 94%.¹

This document serves as a strategic manual for UX architects, detailing where to place elements within non-traditional layouts to maximize "good friction" while avoiding cognitive overload.

1. The Resurgence of Raw: Deconstructing Modern Web Brutalism (2024-2025)

To understand how users scan bold interfaces, we must first define the visual stimuli they are encountering. The resurgence of Brutalism is not merely a stylistic trend; it is a reaction to the "en-shittification" of the web and the blandness of AI-generated content.

1.1 The Cultural Context: From "Brat Summer" to Enterprise UI

The trajectory of web design in 2025 has been heavily influenced by cultural shifts that celebrate the "raw" and "unpolished." We have observed the impact of pop culture phenomena, such as Charli XCX's "brat summer," which popularized a lo-fi, high-contrast aesthetic characterized by fuzzy Arial fonts on lime green backgrounds.² This cultural signal has permeated web design, legitimizing "Modern Web Brutalism" as a valid commercial aesthetic.³

Unlike the architectural brutalism of the 1950s, which was born of necessity and concrete, Modern Web Brutalism is a deliberate choice of excess and function. It is characterized by specific visual markers that defy the "clean" aesthetic of the 2010s:

- **Default System Fonts:** A rejection of custom, polished typography in favor of raw sans-serifs (Arial, Helvetica, Courier) that signal utility over decoration.²
- **High-Contrast Color Blocking:** The use of "dopamine palettes" and solid colors to create distinct visual zones, moving away from soft gradients and shadows.⁴
- **Visible Grids and Borders:** Exposing the skeleton of the website, reminiscent of early HTML tables, but stylized for the modern viewport. This "Anti-Design" approach emphasizes structure over surface polish.⁵
- **Asymmetry and Overlap:** Elements that intentionally overlap or misalign to break the user's expectation of a rigid grid, thereby arresting attention.³

1.2 The "Anti-AI" Aesthetic and Human Authenticity

A critical driver of this trend is the desire for human authenticity. As AI tools increasingly homogenize content, producing glossy but soulless visuals, designers are leaning into "anti-design" to signal human authorship. Research notes a decline in ultra-minimalist, sterile sites in favor of those with distinct editorial voices and "organic, nature-inspired aesthetics" or raw, chaotic energy.³ This shift is not just aesthetic; it is a signal of trust. In an era of deepfakes and algorithmic generation, "rawness" reads as "real." The "stale stock photos" are being replaced by customizable, distinct visuals that convey a unique brand narrative.³

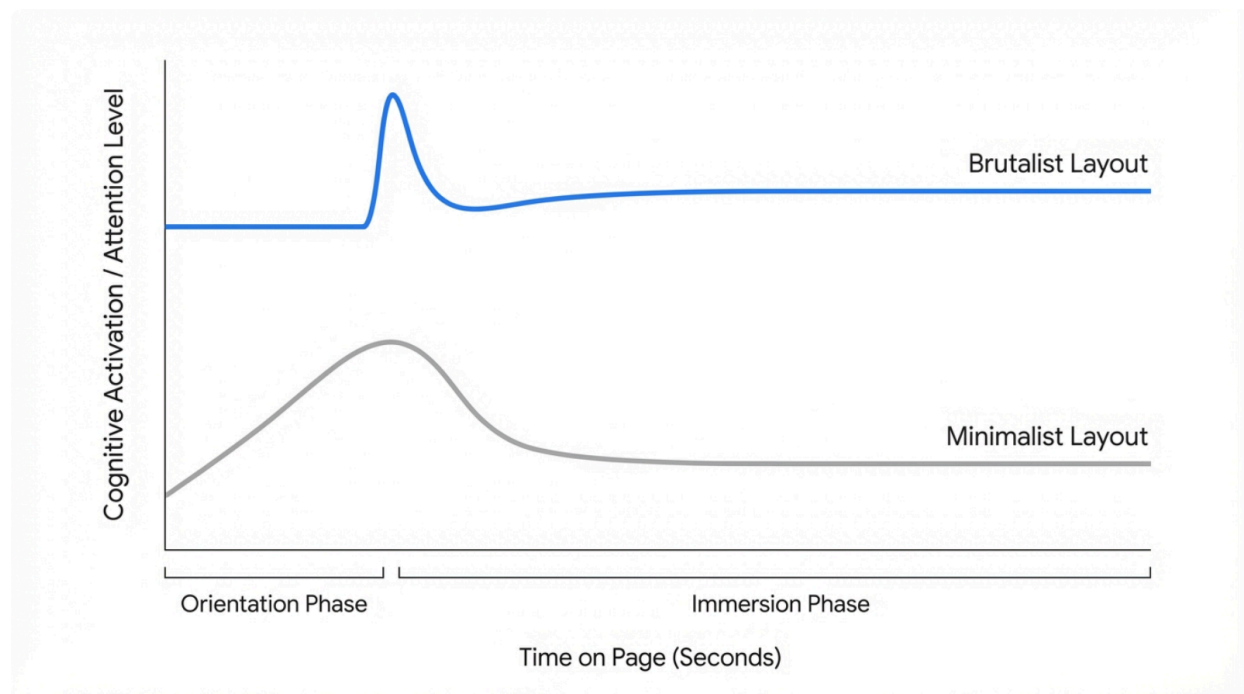
1.3 Cognitive Load and the Brutalist Paradox

The central tension in adopting bold design is **Cognitive Load**. Traditional UX theory, anchored in works like *Don't Make Me Think*, dictates that minimizing cognitive effort is the golden rule. Complex, non-standard layouts ostensibly increase the cognitive effort required to navigate a page because they break established mental models. However, recent observations suggest a paradox: while the *initial* cognitive load is higher, the distinctiveness of the layout creates stronger memory encoding and deeper engagement.⁵

This phenomenon aligns with the "Aesthetic-Usability Effect," where users perceive aesthetically pleasing or striking designs as more usable, even if they inherently possess

higher friction.⁷ The "Pinball Pattern" (discussed in Section 2) suggests that users are willing to expend more attentional energy if the visual payoff is high. The key is ensuring that the friction encountered is "Good Friction"—interaction that slows the user down to appreciate content—rather than "Bad Friction" (confusion).⁸

Cognitive Load vs. Engagement: The Brutalist Paradox



While standard minimalist layouts offer low initial cognitive load, they often suffer from rapid attention decay. Brutalist layouts induce a higher initial 'orientation cost' but can sustain deeper engagement through visual novelty and 'good friction.'

2. Gaze Mechanics in Non-Traditional Layouts

The most significant operational question for designers adopting Avant-Garde layouts is: *How do users actually look at this?* The comfortable certainty of the F-Pattern is eroding as layouts break the grid. Eye-tracking technology has evolved to provide granular insights into these new behaviors, revealing that users adapt their scanning strategies based on the visual hierarchy presented to them.

2.1 The Erosion of the F-Pattern in Avant-Garde Contexts

The F-Pattern, first identified by the Nielsen Norman Group (NN/g) in 2006, describes how users scan text-heavy, homogeneous content: they read the top line, scan down the left, and read a bit less across the middle.¹⁰ This pattern is a symptom of **efficiency** and **lack of visual hierarchy**. Users default to the F-Pattern when a page fails to guide them. It is essentially a "survival mechanism" for processing dense blocks of uniform text.

In 2024-2025, with the rise of broken grids and bold typography, the F-Pattern is often disrupted. When designers use "Spotting" techniques—bold keywords, high-contrast buttons, and asymmetric placement—users switch to alternative patterns. The **"Spotted Pattern"** involves fixating on specific words or chunks that visually stand out (e.g., bold text, colored links), ignoring the rest of the text block.¹³ Similarly, the **"Layer Cake Pattern"** involves scanning headings and subheadings to find relevant sections before diving into the body text.¹³ These patterns demonstrate that bold design elements can effectively override the default F-Pattern behavior, granting the designer control over the user's reading path.

2.2 The Dominance of the "Pinball Pattern" in Broken Grids

For complex, non-linear layouts common in Brutalism, the **"Pinball Pattern"** has emerged as the dominant gaze model. Originally observed in complex Search Engine Results Pages (SERPs) where ads, snippets, and images compete for attention, this pattern is now applicable to Avant-Garde web design.¹⁵

In a Pinball Pattern, the user's gaze bounces non-linearly between highly visual elements.

- **Mechanism:** The eye is drawn to "visual weights"—an oversized image, a neon-colored text block, a 3D element—regardless of its position in the Cartesian grid.¹⁸ The "gravity" of the element determines the gaze path, rather than the reading direction.
- **Implication for Brutalism:** In a broken grid layout, designers cannot rely on the top-left corner for primacy. They must "weight" key conversion elements (CTAs) visually. If a secondary element is too "heavy" (e.g., a massive animated GIF), it will steal the gaze from the primary navigation, causing a "pinball" effect where the user misses the goal.¹⁹
- **Optimization:** To harness the Pinball Pattern, designers must act as "bumpers," using visual hierarchy (size, color, motion) to bounce the eye toward the CTA. The layout must provide a clear "exit velocity" from the chaos toward the conversion point.

2.3 Scrollytelling: The Linear Override and Focused Gaze

While Brutalism creates chaos (Pinball), **Scrollytelling** imposes strict order. Scrollytelling (scrolling + storytelling) utilizes the vertical scroll interaction to trigger animations and content reveals.

- **Linearity vs. Z-Pattern:** Unlike standard pages where users can scan ahead, scrollytelling forces a **linear progression**.²⁰ The user controls the *speed* of time, but not the *sequence* of events. This structure often negates the Z-Pattern, which typically applies to pages with a clear visual hierarchy at the top and bottom but open space in

the middle.¹³

- **Focused Gaze:** Research suggests this format creates a "**Focused Gaze**" or "**Marking Pattern**".¹³ The eye remains relatively fixed in the center (or the active zone) while the content moves *past* the gaze. This behavior is akin to a dancer fixating on an object to maintain balance.
- **Attention Economy:** This format significantly reduces the "lawn-mower" scanning behavior (systematic back-and-forth). It is highly effective for complex narratives because it serializes information, reducing cognitive load by presenting one chunk at a time.²⁰
- **Engagement Metrics:** A case study from the Swiss Federal Statistical Office found that a scrollytelling post achieved the **highest repost rate** and **second highest impression count** among over 400 posts, with an engagement rate of nearly 6%.²⁰ This validates the format's ability to hold attention in an era of waning concentration.

Comparative Gaze Mechanics: F-Pattern vs. Pinball vs. Scrollytelling

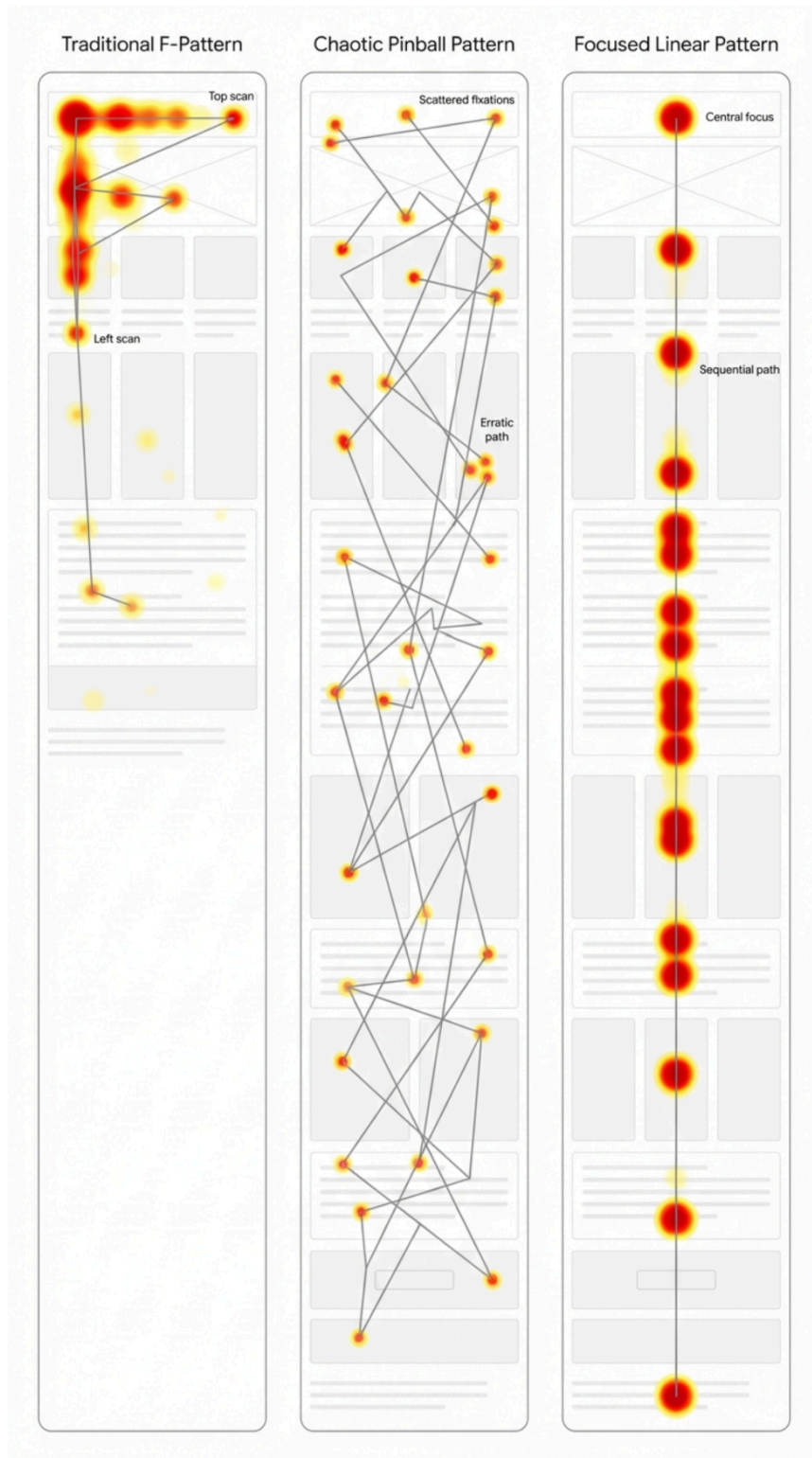


Figure A (Left) shows the efficiency-driven F-Pattern on a standard grid. Figure B (Center) illustrates the 'Pinball Pattern' on a broken grid, where gaze is driven by visual weight. Figure C (Right) demonstrates the 'Focused Linear' path of scrollytelling, where gaze remains centered.

2.4 Comparative Gaze Dynamics

To further elucidate the differences between these patterns, the following table contrasts the primary drivers and user behaviors associated with each layout type.

Gaze Pattern	Triggering Layout	Primary Driver	User Behavior	Best Use Case
F-Pattern	Text-Heavy / Standard Grid	Efficiency / Habits	Scanning top-left to bottom-right; ignoring right side.	Blogs, Articles, Documentation. ¹⁰
Pinball	Broken Grid / Brutalist	Visual Weight	Non-linear bouncing; exploring high-contrast elements.	Brand Awareness, Portfolio, Immersive Experiences. ¹⁵
Layer Cake	Structured / Headings	Information Scent	Scanning only headings to find relevance.	News Sites, Listings, Long-form Content. ¹³
Marking	Scrollytelling	Motion / Centrality	Fixed gaze; content flows through the focal point.	Storytelling, Data Visualization, Complex Explanations. ¹³

3. The Usability vs. Aesthetics Trade-Off: A 2025 Re-evaluation

The debate between "looking good" and "working well" has been fundamentally reframed by the **Aesthetic-Usability Effect**. This psychological phenomenon dictates that users perceive aesthetically pleasing designs as more usable, even if they aren't.⁷ In 2025, this effect is the "shield" that protects bold, brutalist designs from being rejected due to their high friction. It suggests that aesthetics are not merely decoration but a core component of the user's

functional evaluation of a system.

3.1 Interaction Cost and "Good Friction"

Traditional UX aims to reduce friction—minimize clicks, minimize reading, minimize wait times. However, the bold design movement embraces "**Good Friction**".⁸

- **Definition:** Good friction is intentional resistance that forces the user to slow down, consider a decision, or absorb a narrative. It is the difference between a frustrating obstacle and a meaningful pause.
- **Application:** In Avant-Garde layouts, "hover-to-reveal" interactions are a prime example. They hide information (increasing interaction cost) but create curiosity and "micro-rewards" when the user engages.
- **Data on Micro-Interactions:**
 - **Browsing Time Lift:** Google research indicates that dynamic hover interactions (e.g., revealing color options) increase product browsing time by **12%**.²⁴
 - **Satisfaction:** Subtle micro-animations (like a pulse on hover) increase user satisfaction by **17%**.²⁴
 - **Perceived Speed:** Despite adding code weight, interfaces with micro-interactions are perceived as **5% faster** because the immediate visual feedback masks the system's processing time.²⁴
 - **Completion Rates:** Progressive disclosure (a form of friction that hides complexity) can boost form completion by **23%**.²⁴

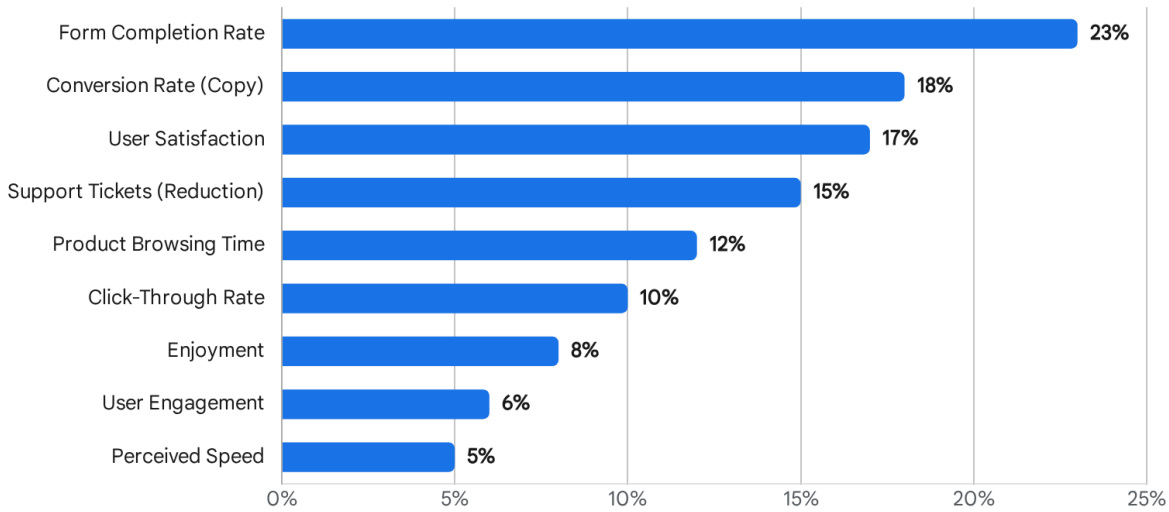
3.2 The ROI of Delight

The investment in "delight" features—often dismissed by engineers as "eye candy"—has a quantifiable Return on Investment (ROI). The data suggests that the "time-on-page" metric, often seen as a negative signal of confusion in utility apps, is a positive signal of **immersion** in brand sites.

- **Click-Through Rate (CTR):** Hover interactions on the Google Play Store increased CTR by **10%**.²⁴
- **Conversion:** Simple changes to micro-copy (e.g., "Get Your Free Trial" vs. "Download") driven by interaction testing lifted conversion by **18%**.²⁴

The ROI of Micro-Interactions: 2024 Performance Metrics

Percentage Lift by Metric



Data from Google research highlights significant performance gains across browsing, satisfaction, and conversion metrics when micro-interactions are effectively implemented.

Data sources: [Medium \(Mustafa Jamal Nasser\)](#).

3.3 The Frustration Threshold and Accessibility

While the Aesthetic-Usability Effect is powerful, it has a breaking point. If the interaction cost exceeds the user's motivation or ability, the effect collapses, and users bounce.

- **Cognitive Overload:** If the "pinballing" required to find a CTA exceeds the user's motivation, they will bounce. The "Pinball Pattern" is only successful if the visual weights eventually guide the eye to the goal.
 - **Accessibility Risks:** Modern Brutalism's reliance on raw text and high contrast *can* be good for accessibility (WCAG compliance), but its chaotic layouts can be a nightmare for screen readers and keyboard navigation. Snippet ³ warns that 95.9% of homepages fail compliance checks. A "bold" design that fails accessibility is not avant-garde; it is exclusionary. High-contrast modes and raw text are beneficial, but overlapping elements must be coded with strict DOM ordering to ensure screen readers traverse the content logically.
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4. The 3D UI Frontier: Engagement Mechanics

Moving beyond 2D layouts, the integration of 3D elements (using libraries like Spline, Three.js, or React Three Fiber) represents the next frontier of "bold" design. This is no longer about static aesthetics but about **interactive dimensionality**.

4.1 Engagement & Conversion Statistics

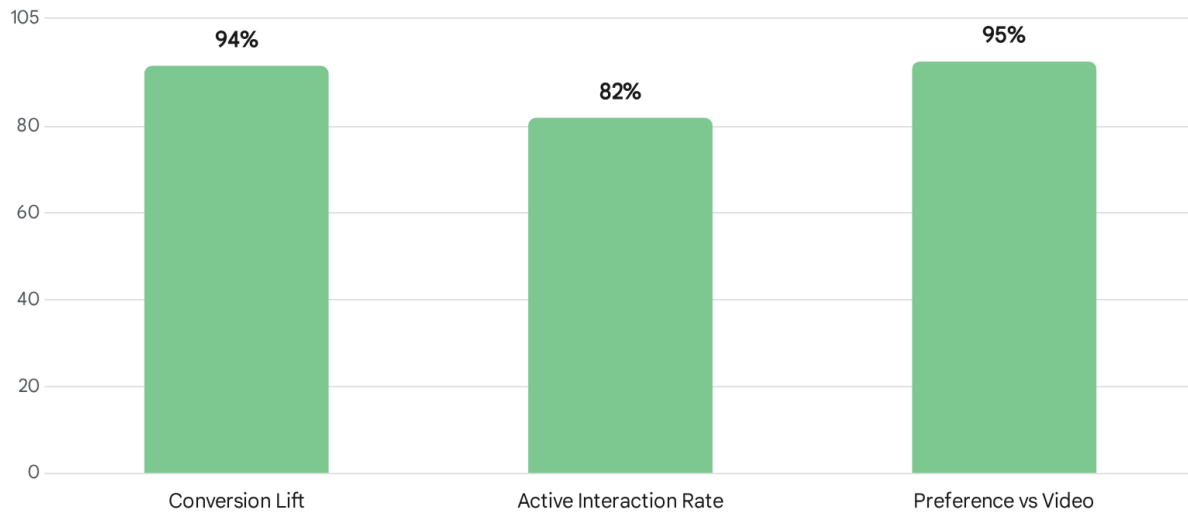
The data on 3D integration is overwhelmingly positive, particularly for e-commerce and product showcases. The ability to manipulate an object in 3D space transforms the user from a passive observer to an active participant.

- **Conversion Lift:** Adding 3D content to product pages results in a staggering **94% increase in conversion**.¹
- **Interaction Rates:** **82%** of viewers who see a 3D asset will activate and interact with it.¹ This is a massive engagement signal compared to passive video consumption.
- **Dwell Time:** **34%** of users interact with 3D content for more than **30 seconds**.¹ In the attention economy, 30 seconds is an eternity.
- **Preference:** **95%** of users prefer interactive 3D content over static video playback.¹

3D vs. Static: The Engagement Gap (2025)

KEY PERFORMANCE METRICS (%)

● 3D Content Performance



3D content significantly outperforms static media, driving a 94% lift in conversion and achieving a 95% user preference rate over video.

Data sources: [CGI Backgrounds](#)

4.2 The Psychology of 3D Interaction

Why does 3D work so well? Several psychological factors are at play:

1. **Agency and Control:** Unlike video, which is passive, 3D requires user input (rotation, zoom). This active participation creates a sense of ownership, known as the **Endowment Effect**—users value things more when they feel they "own" or control them.
2. **Reduced Information Asymmetry:** 66% of shoppers say a 3D configurator increases their buying confidence.¹ Being able to inspect a product from all angles reduces the risk associated with online shopping, bridging the gap between digital and physical retail.
3. **Novelty and Orientation:** In a sea of flat 2D cards, a 3D element breaks the depth plane, triggering an orienting response in the brain. It signals that the object is "real" and worthy of attention.

4.3 Technical Implementation & Performance

The trade-off for 3D is **Performance Cost** (load times). However, advances in WebGL and optimized libraries like Spline have minimized this impact. The key is **"Judicious Integration"**.²⁵ 3D elements should be used as "Hero" elements or specific product

interactables, not as decorative background noise that clogs the main thread. Furthermore, ensuring a high-quality static fallback is essential for users on low-power devices to avoid the "Bad Friction" of a loading spinner.

5. Strategic Recommendations: Where to Place Elements

Based on the synthesis of "Pinball" gaze patterns, "Good Friction," and 3D engagement data, we propose the following spatial strategies for Bold/Avant-Garde layouts. These recommendations aim to balance the aesthetic impact with functional usability.

5.1 The "Anchor & Orbit" Strategy for Broken Grids

In a chaotic/brutalist layout, you cannot rely on the F-Pattern to guide the user to the Call to Action (CTA). Instead, you must engineer visual gravity.

- **The Anchor:** Place your primary CTA in a **High-Contrast Container** (e.g., a black button on a lime green field). This acts as the "black hole" of the layout—the point of highest visual gravity. It should be the heaviest element on the page in terms of color saturation or size.
- **The Orbit:** Distribute secondary content (text, decorative images, marquees) around this anchor. The user's eye will "pinball" off the secondary elements but, due to visual gravity, will inevitably be pulled back to the high-contrast Anchor.¹⁸
- **Avoid:** Placing the CTA in a "blind spot" (e.g., bottom right) without strong visual vectors pointing to it. The bottom-right is often ignored in F-Patterns, but in Pinball patterns, it can be effective *only* if a visual line (like a diagonal graphic) leads the eye there.

5.2 The "Visual Rhythm" for Scrollytelling

To prevent the "lawn-mower" fatigue (constant left-right scanning) in scrollytelling:

- **Center-Stage Focus:** Keep the active narrative element in the center 50% of the viewport.²⁶ Eye-tracking studies confirm that users focus mainly on the center of the screen in these layouts.
- **Dynamic Backgrounds:** Use the background for context (3D scenes, maps) while the foreground text remains static or fades in/out. This separates the "Atmosphere" (peripheral vision) from the "Information" (foveal vision), reducing cognitive load.
- **Progress Indicators:** Always include a visual progress bar or chapter markers. In a linear format, users need to know "how much is left" to maintain motivation and orientation.²⁰

5.3 Micro-Interaction Zones

- **Hover-Reveal:** Use for **Product Details** (colors, sizes, price breakdowns). Do NOT use

for critical navigation (users shouldn't have to hunt for the menu). The interaction cost is justified by the "reveal" of detailed data, but not for basic wayfinding.

- **3D Triggers:** Place interactive 3D elements "Above the Fold" (or in the first scroll depth) to capture initial attention and increase time-on-page immediately. This sets the hook for the engagement session.

5.4 Typography and Attention

The psychology of typography plays a crucial role in these layouts.

- **Bold Typography:** Research confirms that bold fonts are associated with strength and importance, triggering immediate attention.²⁷ They act as "stops" in the pinball gaze, arresting the eye's movement.
- **Sans Serif Dominance:** The use of simple sans-serif fonts (Arial, Helvetica) in Brutalism² aids readability by reducing the visual noise of the letters themselves, allowing the *layout* to be the source of complexity.

Conclusion: The Era of "Functional Chaos"

The UX landscape of 2025 is defined by a move away from safe, invisible design toward **Functional Chaos**. "Bold" design—whether through Brutalist typography, broken grids, or immersive 3D—is not just an aesthetic choice; it is a strategy for **attention capture** in a saturated digital environment.

The data proves that users will forgive, and even enjoy, higher interaction costs (friction) if the reward is novelty, agency, and authenticity. By understanding the **Pinball Pattern** and leveraging the **Aesthetic-Usability Effect**, designers can build interfaces that are unapologetically loud yet highly effective. The key lies in the rigorous application of visual hierarchy: if you break the grid, you must build a stronger magnet.

Key Takeaways for the Practitioner

1. **Embrace the Pinball:** In non-traditional layouts, design for *visual weight*, not *Cartesian position*. Guide the eye with contrast, not coordinates.
2. **Invest in Good Friction:** Use micro-interactions to create "speed bumps" that increase immersion and perceived value.
3. **Go 3D for Commerce:** The 94% conversion lift¹ is too significant to ignore for product-based sites.
4. **Test for "Bad" Friction:** Monitor "Rage Taps"²⁸ and bounce rates. If the boldness blocks the goal, it's not Brutalism; it's broken.
5. **Prioritize Accessibility:** Ensure that underneath the chaos, the DOM structure remains logical for assistive technologies. Authenticity should not come at the cost of inclusion.

Works cited

1. 16 Key Stats About 3D Content and Ecommerce for 2025 - CGI.Backgrounds, accessed on January 9, 2026, <https://www.cgibackgrounds.com/blog/16-key-stats-about-3d-content-and-ecommerce-for-2025>
2. e9digital Checklist: Make Your Website Look 10 Years Younger With 9 Design Trends in 2025, accessed on January 9, 2026, <https://e9digital.com/website-trends/>
3. 2025 Trends Forecast - Fyresite's Predictions, accessed on January 9, 2026, <https://www.fyresite.com/2025-trends-forecast-fyresites-predictions/>
4. 20 Top Web Design Trends 2026 | TheeDigital, accessed on January 9, 2026, <https://www.theedigital.com/blog/web-design-trends>
5. Neo Brutalism UI Design Trend for a Bold and Impactful User Experience - Onething Design, accessed on January 9, 2026, <https://www.onething.design/post/neo-brutalism-ui-design-trend>
6. Web Design Embraces the Anti-Design Trend - Graticle Design, accessed on January 9, 2026, <https://graticle.com/blog/web-design-embraces-the-anti-design-trend/>
7. How do aesthetics impact usability in UX: The aesthetic-usability effect - LogRocket Blog, accessed on January 9, 2026, <https://blog.logrocket.com/ux-design/aesthetic-usability-effect-ux/>
8. Do this one thing well to power-up your PLG motion, accessed on January 9, 2026, <https://www.plg.news/p/friction-logs>
9. The Good, the Bad and the Ugly: A Story About Design and Friction - Eleken, accessed on January 9, 2026, <https://www.eleken.co/blog-posts/a-story-about-design-and-friction>
10. How Users Scan: Learning from Eye Tracking Research | Codecademy, accessed on January 9, 2026, <https://www.codecademy.com/article/how-users-scan>
11. Explained: the F-shape pattern for reading content - Writeful, accessed on January 9, 2026, <https://writefulcopy.com/blog/f-shaped-pattern-explained>
12. F-Shaped Pattern for Reading Content - Nick Babich, accessed on January 9, 2026, <https://babich.biz/blog/f-shaped-pattern-for-reading-content/>
13. Eye-Tracking Research & Patterns in Digital Reading | by CSNA MEDIA | Medium, accessed on January 9, 2026, <https://medium.com/@CSNA-MEDIA/eye-tracking-reserch-patterns-in-digital-reading-fe6aa5f14eed>
14. Notes on Information Scanning Pattern (IA : 3) : r/UXDesign - Reddit, accessed on January 9, 2026, https://www.reddit.com/r/UXDesign/comments/tds3n7/notes_on_information_scanning_pattern_ia_3/
15. frutik/awesome-search: Awesome Search - this is all about the (e-commerce, but not only) search and its awesomeness - GitHub, accessed on January 9, 2026, <https://github.com/frutik/awesome-search>
16. Susanne Lumsden, Author at Henley Hall Press, accessed on January 9, 2026,

- <https://henleyhallpress.co.uk/author/susanne/>
17. Rethinking eye-tracking: Patterns with purpose | by Ux Spotlight - Medium, accessed on January 9, 2026, <https://medium.com/@ux.spotlight/rethinking-eye-tracking-patterns-with-purpose-e9621cc3b834>
 18. SEO Ranking Reports: Accounting For Search Visibility Changes - seoClarity, accessed on January 9, 2026, <https://www.seoclarity.net/blog/seo-rank-report-what-to-include>
 19. Inclusive Design Patterns 2025 | PDF | Accessibility - Scribd, accessed on January 9, 2026, <https://www.scribd.com/document/824140223/Inclusive-Design-Patterns-2025>
 20. From storytelling to scrollytelling – modern digital ... - UNECE, accessed on January 9, 2026, https://unece.org/sites/default/files/2023-08/DissComm2023_S3_Switzerland_Ghanfili_Paper.pdf
 21. Visual Hierarchy: Organizing content to follow natural eye movement patterns | IxDF, accessed on January 9, 2026, <https://www.interaction-design.org/literature/article/visual-hierarchy-organizing-content-to-follow-natural-eye-movement-patterns>
 22. Understanding the Aesthetic Usability Effect: Design's Psychological Impact - Claritee, accessed on January 9, 2026, <https://claritee.io/blog/understanding-the-aesthetic-usability-effect-designs-psychological-impact/>
 23. What is Friction? Digital Marketing Hidden Killer - Arfadia, accessed on January 9, 2026, <https://www.arfadia.com/glossary/EN/friction>
 24. Captivate Users: Unveiling the Power of Micro-Interactions in 2024 | by Mustafa Jamal Nasser | Medium, accessed on January 9, 2026, <https://medium.com/@MustafaJamalNasser/captivate-users-unveiling-the-power-of-micro-interactions-in-2024-4f5301c74c39>
 25. Future of Web Design: Emerging Trends for 2025 - Digital Content Labs, accessed on January 9, 2026, <https://www.digitalcontentlabs.co.uk/2024/01/15/emerging-trends-in-web-design-2025/>
 26. Improvements to Visual Design on the Online News Portal ABC.com by Employing Usability and Eye Tracking Methods - MDPI, accessed on January 9, 2026, <https://www.mdpi.com/2673-4591/84/1/72>
 27. The Psychology of Font Choice: How Typography Impacts Content Engagement, accessed on January 9, 2026, <https://www.thewritersforhire.com/the-psychology-of-font-choice-how-typography-impacts-content-engagement/>
 28. User Engagement Metrics - The Complete Guide 2025 - UXCam, accessed on January 9, 2026, <https://uxcam.com/blog/user-engagement-metrics/>